CURRICULUM VITAE

Ankit Mathanker
Ph.D. Candidate
Dept. of Chemical Engineering (Fall 2021 -)
University of Michigan, Ann Arbor, USA
Email: ankitma@umich.edu

Academic Qualifications

Master of Science (Chemical Engineering) — *University of Alberta, Canada*
GPA: 3.8/ 4.0 — *Jan. 2018 – Jan. 2020*

Bachelor of Technology (Chemical Engineering) — *IIT (ISM) Dhanbad, India*
GPA: 8.95/ 10.0 — *July 2013 – May 2017*

Research Experience

1. *Research Assistant* — *University of Alberta, Canada*
   Oct. 2020 – March 2021
   - Process parameter *optimization* for hydrothermal liquefaction
   - *Data collection* and *feature extraction* for various feedstocks and process parameters
   - *Implementing machine learning* concepts for process optimization

2. *Graduate Research Assistant* — *University of Alberta, Canada*
   Hydrothermal treatment of biomass to produce biofuels. — *Jan. 2018 – Jan. 2020*
   - *Initiated* first project on biofuels in coal technology group
   - *Investigated* and *optimized* process in autoclave reactor
   - *Trouble-shot* and *developed* GC-MS operation procedure
   - *Characterized* products on GC-MS, GC-FID, CHNS, FTIR, SEM, BET and TGA
   - Developed SOP and conducted *safety inspections* meetings

3. *Research Assistant (SRFP-Indian Academy of Science 2016)* — *IIT Guwahati, India*
   Synthesis and application of low-cost tubular ceramic membrane (TCM). — *May - July 2016*
   - *Synthesised* TCM through extrusion process
   - Measured water flux and porosity using experimental and analytical skills
   - Conducted wastewater treatment experiments on TCM and performed spectroscopy to check filtration

4. *Undergraduate Dissertation* — *IIT (ISM) Dhanbad, India*
   Synthesis of fly ash-based PEI impregnated adsorbents for CO_2 capture studies. — *July 2016 - April 2017*
   - *Synthesised* fly ash and SBA-15 based adsorbents in laboratory
   - Performed *activation and impregnation* of adsorbents
   - *Characterized* adsorbents on BET, TGA and SEM

   Modelling and Simulation of CO_2 adsorption in a packed bed. — *May 2015 - August 2015*
   - *Modelled and simulated* adsorbent packed bed and compared different breakthrough curves on Aspen Adsorption v8.6.

Publications

1. **Mathanker, A.,** Pudasainee, D., Kumar, A., Gupta, R., *Hydrothermal liquefaction of lignocellulosic biomass feedstock to produce biofuels: Parameter study and products characterization.* *Fuel 271 (2020) 117534*.
2. **Mathanker, A.,** *Hydrothermal liquefaction of lignocellulosic biomass to produce biofuels.*(*Thesis 2020*).
3. **Mathanker, A.,** Das, S., Pudasainee, D., Khan, M., Gupta, R., *A review of hydrothermal liquefaction of biomass for biofuels production with a special focus on the effect of process parameters, co-solvents, and extraction solvents.* *Energies 14 (16), 4916*.

Conference Presentations

1. *Clearwater Clean Energy Conference:44th International Conference on Clean Energy* — *Florida, USA*
   Mathanker, A., Pudasainee, D., Kumar, A., Gupta, R., (2019). *Parametric study on hydrothermal treatment of biomass to produce biofuels.*
2. *Faculty of Engineering Graduate Research Symposium* — *University of Alberta, Canada*
   Mathanker, A., Javed, K., Pudasainee, D., Kumar, A., Gupta, R. (2018). *Study of hydrothermal treatment of corn stover to produce biofuels.*

3. *Future Energy System, Open House Symposium* *University of Alberta, Canada*
   Vaezi, M., Javed, K., Pudasainee, D., **Mathanker, A.**, Kumar, A., Gupta, R. (2018). *Integration of pipeline hydro-transport and hydrothermal conversion technologies to produce biofuels.*
4. Falling Wall Lab, 5th Annual Event *University of Alberta, Canada*
   Mathanker, A. 2018. *Breaking the wall of energy crisis and pollution.*
5. 3 Minute Thesis, *University of Alberta, Canada*
   Mathanker, A. *(2018). Fueling the future.*
6. *CHEMCON 2016-69th Annual Session of Indian Institute of Chemical Engineers* *Anna University, India*
   Mathanker, A., Kumar, M., Kumar, V., Pugazhenthi, G. (2016). *Effect of sintering temperature on characteristics of low-cost tubular ceramic support manufactured from Indian raw materials for membrane applications.*
7. *CHEMCON 2015-68th Annual Session of Indian Institute of Chemical Engineers* *IIT Guwahati, India*
   Mathanker, A., Dey, R., Samanta, A. (2015). *Simulation of CO_2 adsorption in a packed bed column using Aspen Adsorption.*

Teaching Experience

1. **Graduate Teaching Assistant** *University of Alberta, Canada*
   CHE 316- Equilibrium stage processes, Spring and Fall term. *May 2019 - December 2019*
   - *Taught* seminars for more than 40 hours to a class of 48 stuends.
   - *Prepared* assignments, seminars and quizzes and enhanced *teaching and presentation skills*
2. **Graduate Teaching and Learning (**Foundation and Practicum) *University of Alberta, Canada*
   - Drafted *teaching philosophy, learning objectives, lesson planning* strategies
   - Harnessed skills in *microteaching, classroom management* and *leadership skills.*

Skills

- Instrumentation expertise: GC-MS, TGA, SEM, BET, FTIR, Spectroscopy (All advance)
- Programming Skills: Python (Intermediate), MATLAB (Intermediate), Machine Learning (Intermediate)
- Software/ Editor Knowledge: LaTeX, Jupyter Lab, AspenTech, Microsoft Office Tools (Advance)
- Professional development courses:
  - Introduction to Machine Learning (Audited) – (In class – University of Alberta)
  - Python data structure (Coursera – University of Michigan)
  - Mathematics for machine learning Specialization (Coursera – Imperial College London)
  - Applied machine learning – Specialization (Coursera – University of Michigan)

Achievements

1. Captain Thomas Farrell Greenhalgh Memorial Graduate Scholarship *University of Alberta, Canada*
2. Mary Louise Imrie Graduate Student Award *University of Alberta, Canada*
3. Graduate Student Association Travel Award *University of Alberta, Canada*
4. Summer Research Fellowship Programme Scholarship *Indian Academy of Science, Bengaluru, India*

Volunteer Experience

1. Vice President**,** Toastmasters International *University of Alberta (2018- 2019)*
   Managed memberships, public relations and conducted public speaking competitions.
2. Volunteer, UNITEA, Community Social Work Team *University of Alberta (2018- 2019)*
   Promoting mental health awareness
3. Volunteer, Kartavya, India *IIT (ISM) Dhanbad (2014-2016)*
   NGO/NPO focusing on rural education and health issues